

# Hamza Waqas

London, UK | [hamzawaqas2005@gmail.com](mailto:hamzawaqas2005@gmail.com) | [GitHub](#) | [Portfolio](#) | [LinkedIn](#) | +44 749 039 4106

## Education

City St George's, University of London  
BEng Aerospace Engineering (*Predicted First Class with Honours*)

London, UK  
Graduation July 2027

## Experience

### Panasonic Avionics Corporation

Summer Work Experience

July 2025

London, UK

- Looked into integration of IFE systems onto Airbus aircraft and the current issues they had on the customer aircraft.
- Followed tests taken to recreate the power interrupts from switching from APU to GPU or to engine power.
- Studied older IFE systems facing issues to draw parallels and determine if these were hardware or aircraft related.
- Shadowed the engineers working on the project, covering part of the engineering investigations and execution of the aircraft test plan.

### Formula Student UK

Head of Mechanical Systems

September 2024 – Present

Extracurricular in 1<sup>st</sup> and 2<sup>nd</sup> year

- Led chassis design and structural validation for the team's first EV and car in 8 years, targeting the 2027 season.
- Designed the full front and rear suspension geometry from a clean-sheet chassis, defining roll centre height, camber, anti-dive, and pushrod/bellcrank motion ratio targets.
- Validated structural integrity to 40g frontal/side impact loading while cutting chassis mass from 57.8kg to 53.2kg (safety factor >1.6); delivered full vehicle design-to-build in 10 months against fixed facility constraints.

## Projects

### CAGE (Confined Aerial Geometry) - AEW&C system for UAVs

- Designed the complete software and firmware architecture for a radar sensing system built from four independent units across dual ESP32-S3 controllers, with C++ firmware handling sensor acquisition and live UDP telemetry over Wi-Fi.
- Built a Python fusion and visualization pipeline that ingests live UDP streams from multiple sensors, merges detections, and renders a unified 360° polar display with overlapping coverage in real time.
- Implemented and tested the full pipeline from sensor to display on a standalone rig, validating telemetry reliability, detection coverage, and tracking of multiple targets ahead of integration.

### ML - Based Gas Turbine Emissions Modelling

- Built a complete Python machine learning pipeline (scikit-learn) predicting gas turbine CO and NOx emissions from a 35,499-row sensor dataset, including data cleaning (missing values, duplicates, IQR based outlier removal) and feature engineering (humidity-temperature ratio, temperature-pressure interaction) informed by correlation analysis.
- Trained and compared four models, polynomial, ridge, and lasso regression plus a tuned gradient boosting regressor, evaluated with MAE, RMSE, and R<sup>2</sup> on a fixed train, validation, and test split, with gradient boosting performing best.
- Achieved a held-out test R<sup>2</sup> of 0.71 for CO and 0.65 for NOx, built the matplotlib visualizations (heatmaps, boxplots, predicted versus actual plots) used to interpret features and validate performance, and authored the full technical report.

### Spacecraft ADCS Simulator (Attitude Determination and Control System)

- Built a full 6-DOF simulation of a spacecraft's attitude dynamics and control system in MATLAB from first principles, modelling sensor, actuator, and disturbance behaviour as interchangeable modules, validated to approximately 10<sup>-13</sup> relative accuracy against closed-form solutions.
- Implemented three interchangeable attitude controllers (PID, LQR, LQI), building the LQR/LQI Riccati solver from scratch and verifying it to 2x10<sup>-10</sup> agreement against MATLAB's native solver and independently derived solutions.
- Designed a Monte Carlo validated Kalman filter estimator and a supervisory state machine (SAFE, ACQUIRE, NOMINAL) for autonomous fault recovery, quantifying a 4.2x pointing error increase under realistic uncertainty and recovering performance within approximately 260 seconds without mode chattering.
- Currently porting over to python (open sourced) with an extended suite of features to make it more attainable and education focused for those interested in spacecraft technology.

## Skills

**Languages:** Python (scikit-learn, Pandas, NumPy, Matplotlib), C/C++ (embedded systems, Arduino/ESP32), MATLAB

**Tools and Technologies:** Git, VS Code, Ardupilot, Mavlink, SolidWorks/FEA, ANSYS, ABAQUS, SIEMENS NX/FEA, Fusion 360, XFLR5, 3D Printing, Microsoft Office, Google Suite